



VBF-T7R3-DVPR-01 Case t7_r3 | 2026-08-26

Prepared: _____ Reviewed: _____ Approved: _____

Virtual design package - simulation caliber (no physical build)

VBF Design Verification Plan & Report (virtual-test version) - dvpr

****Document number**:** VBF-T7R3-DVPR-01 · ****Case**:** t7_r3 · ****Date**:** 2026-08-26

All result values are mechanically taken from bda simulation outputs. Prepared / Reviewed / Approved: _____

#	Verification item	Condition	Result value	Determination	Source
1	1C discharge capacity	6.28 A CC to 2.5 V, 25 °C (SPMe)	6.2569 Ah	[PASS] vs nominal 6.28 Ah	`cell/r7_v19_final_1c.json:capacity_ah`
2	1C discharge capacity (DFN cross-check)	6.28 A CC to 2.5 V (DFN)	6.28 Ah	[PASS] consistent	`cell/r6_v19_1c_dfn.json:capacity_ah`
3	Energy density	contract formula $\text{intV} \cdot \text{I} \cdot \text{C} \cdot \text{dt} \div \text{mass}$	482.43 Wh/kg	≥ 327.18 [PASS] PASS	`cell/r7_v19_final_energy.json:energy_density_wh_kg`
4	4C fast-charge temperature rise	25.12 A charge from 2.5 V to 4.2 V, 45 °C, lumped thermal	T_max 363.2 K (90.1 °C)	[PASS] no criterion on T; recorded	`cell/r7_v19_final_4c45c.json:T_max_K`
5	4C fast-charge lithium plating	same run, `--plating`, DFN anode surface potential min	$\geq 0.0171 \text{ V}$	plated = false [PASS] PASS	`cell/r7_v19_final_4c45c.json:anode_potential_v`
6	SEI growth durability	aging 1C $\times 100$ cyc @ 45 °C (SEI ec-reaction-limited, isothermal)	SEI end 512.92 nm	$\leq 550 \text{ nm}$ [PASS] PASS	`cell/r7_v19_final_aging45c.json:sei_thickness_nm_end`
7	Nail penetration thermal runaway (virtual ODE)	q_nail = 10 W, mcp = 41.94 J/K (46.62 g $\times$ 900), hA = 0.5 W/K, T _■ = 298.15 K	triggered = false, T_max = 318.2 K, dT/dt_max = 0.236 K/s	[PASS] PASS	`cell/r7_v19_final_nail_hA05.json`
8	Nail sweep - cooling margin	hA = 0.05 / 0.2 / 0.3 / 0.4 / 0.5 W/K	triggers at 346 s / triggers at 2097 s (405 K) / pass 331.7 K / pass 323.2 K / pass 318.2 K	min required hA ~ 0.3 W/K -> design 0.5 W/K	`cell/r7_v19_final_nail_hA*.json`
9	Voltage window	parameter set cut-offs	2.5 - 4.2 V	[PASS] Chen2020 set	

****Not covered (N/A - beyond pure simulation boundary, requires physical experiment)**:**

overcharge-to-thermal-runaway physical test, crush, drop, vibration, calendar life, EOL capacity retention, low-temperature (< 0 °C) performance, rate-pulse internal resistance, production tolerances. (Virtual overcharge protocol exists but was not in the task criteria.)

****Conclusion**:** All six task criteria verified PASS in the virtual battery (ED 482.4 $\geq$ 327.18 Wh/kg; SEI 512.9 $\leq$ 550 nm; plated = false at 4C/25.12 A; triggered = false at 10 W nail with designed cooling). Uncovered items above are listed for release-book honesty; they require physical prototypes.